--> declare(g, constant);

(% o1) done

--> declare(Ω, constant);

(% o2) done

--> declare(φ, constant);

(% o3) done

--> assume(φ >= - %pi / 2, φ <= %pi / 2, g > 0, Ω > 0, t >= 0);

(% o4) $\left[-\phi-\frac{\pi}{2}\leq 0\,,\phi-\frac{\pi}{2}\leq 0\,,0<g\,,0<\Omega\,,t\geq 0\right]$

--> eq1: 2 * Ω * sin(φ) * diff(y(t), t, 1) - 2 * Ω * cos(φ) * diff(z(t), t, 1) = diff(x(t), t, 2);

(eq1) $2\Omega\sin\left(\phi\right)\left(\frac{d}{dt}\,y(t)\right)-2\Omega\cos\left(\phi\right)\left(\frac{d}{dt}\,z(t)\right)=\frac{d^2}{dt^2}\,x(t)$

--> eq2: - 2 * Ω * sin(φ) * diff(x(t), t, 1) = diff(y(t), t, 2);

(eq2) $-\left(2\Omega\sin\left(\phi\right)\left(\frac{d}{dt}\,x(t)\right)\right)=\frac{d^2}{dt^2}\,y(t)$

--> eq3: - g + 2 * Ω * cos(φ) * diff(x(t), t, 1) = diff(z(t), t, 2);

(eq3) $2\Omega\cos\left(\phi\right)\left(\frac{d}{dt}\,x(t)\right)-g=\frac{d^2}{dt^2}\,z(t)$

--> sol: desolve([eq1, eq2, eq3], [x(t), y(t), z(t)]);

"Is "

$$4\Omega^2\sin\left(\phi\right)^2+4\Omega^2\cos\left(\phi\right)^2$$

" positive or zero?"

positive;

(sol)

$$[x(t) = -\left(\frac{\cos(\phi)\left(\frac{d}{dt}z(t)\Big|_{t=0}\right) - \sin(\phi)\left(\frac{d}{dt}y(t)\Big|_{t=0}\right) - 2\Omega\, x(0)\sin(\phi)^2 - 2\Omega\, x(0)\cos(\phi)^2}{2\Omega\sin(\phi)^2 + 2\Omega\cos(\phi)^2}\right) + \frac{\cos\left(2\Omega\sqrt{\sin(\phi}\right.}{}$$

```
-->    sol: trigsimp(sol);
```

(sol)

$$[x(t) = (\,(2\Omega\cos(\phi)\cos(2\Omega t) - 2\Omega\cos(\phi))\left(\frac{d}{dt}z(t)\Big|_{t=0}\right) + (2\Omega\sin(\phi) - 2\Omega\sin(\phi)\cos(2\Omega t))\left(\frac{d}{dt}y(t)\Big|_{t=0}\right) +$$

```
-->    sol: factor(sol);
```

(sol)

$$[x(t) = (\,2\Omega\cos(\phi)\cos(2\Omega t)\left(\frac{d}{dt}z(t)\Big|_{t=0}\right) - 2\Omega\cos(\phi)\left(\frac{d}{dt}z(t)\Big|_{t=0}\right) - 2\Omega\sin(\phi)\cos(2\Omega t)\left(\frac{d}{dt}y(t)\Big|_{t=0}\right) +$$

```
-->    x_0: at(x(t), t = 0)$
       y_0: at(y(t), t = 0)$
       z_0: at(z(t), t = 0)$
       dx_0: at(diff(x(t), t, 1), t = 0)$
       dy_0: at(diff(y(t), t, 1), t = 0)$
       dz_0: at(diff(z(t), t, 1), t = 0)$

-->    declare(h, constant)$
       assume(h > 0)$;

-->    sol_tower: factor(subst([
       x_0 = 0,
       y_0 = 0,
       z_0 = h,
       dx_0 = 0,
       dy_0 = 0,
       dz_0 = 0],
       sol));
```

(sol_ tower)

$$[x(t) = -\left(\frac{g\cos(\phi)(\sin(2\Omega t) - 2\Omega t)}{4\Omega^2}\right), y(t) = -\left(\frac{g\cos(\phi)\sin(\phi)\left(\cos(2\Omega t) + 2\Omega^2 t^2 - 1\right)}{4\Omega^2}\right), z(t) = -\left(\frac{g\sin(\phi}{}\right.$$

```
-->    expand(subst(φ = 0, sol_tower));
```

(% o20) $\left[x(t) = \frac{gt}{2\Omega} - \frac{g\sin(2\Omega t)}{4\Omega^2}, y(t) = 0, z(t) = \frac{g\cos(2\Omega t)}{4\Omega^2} - \frac{g}{4\Omega^2} + h\right]$

-->
```
expand(subst(φ = %pi / 2, sol_tower));
```

(% o21) $\left[x(t)=0\,,y(t)=0\,,z(t)=h-\frac{g t^2}{2}\right]$

-->
```
expand(subst(φ = - %pi / 2, sol_tower));
```

(% o22) $\left[x(t)=0\,,y(t)=0\,,z(t)=h-\frac{g t^2}{2}\right]$

-->
```
declare(v_0, constant)$
declare(θ, constant)$
assume(v_0 > 0, θ > 0, θ < %pi / 2)$;
```

-->
```
sol_east_launch: factor(subst([
x_0 = 0,
y_0 = 0,
z_0 = 0,
dx_0 = v_0 * cos(θ),
dy_0 = 0,
dz_0 = v_0 * sin(θ)],
sol));
```

(sol_ east_ launch)

$$[\,x(t)=-\left(\frac{g\cos(\phi)\sin(2\Omega t)-2v_0\Omega\cos(\theta)\sin(2\Omega t)-2v_0\Omega\sin(\theta)\cos(\phi)\cos(2\Omega t)-2g\Omega\cos(\phi)t+2v_0\Omega\sin(\theta)}{4\Omega^2}\right.$$

-->
```
expand(subst(φ = 0, sol_east_launch));
```

(% o27)

$$[\,x(t)=\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}-\frac{g\sin(2\Omega t)}{4\Omega^2}+\frac{v_0\sin(\theta)\cos(2\Omega t)}{2\Omega}+\frac{gt}{2\Omega}-\frac{v_0\sin(\theta)}{2\Omega}\,,y(t)=0\,,z(t)=\frac{v_0\sin(\theta)\sin(2\Omega}{2\Omega}$$

-->
```
expand(subst(φ = %pi / 2, sol_east_launch));
```

(% o28)

$$\left[x(t)=\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\,,y(t)=\frac{v_0\cos(\theta)\cos(2\Omega t)}{2\Omega}-\frac{v_0\cos(\theta)}{2\Omega}\,,z(t)=v_0\sin(\theta)t-\frac{g t^2}{2}\right]$$

-->
```
expand(subst(φ = - %pi / 2, sol_east_launch));
```

(% o29)

$$\left[x(t) = \frac{v_0 \cos(\theta) \sin(2\Omega t)}{2\Omega}, y(t) = \frac{v_0 \cos(\theta)}{2\Omega} - \frac{v_0 \cos(\theta) \cos(2\Omega t)}{2\Omega}, z(t) = v_0 \sin(\theta) t - \frac{g t^2}{2}\right]$$

```
-->     sol_north_launch: factor(subst([
        x_0 = 0,
        y_0 = 0,
        z_0 = 0,
        dx_0 = 0,
        dy_0 = v_0 * cos(θ),
        dz_0 = v_0 * sin(θ)],
        sol));
```

(sol_ north_ launch)

$$[x(t) = -\left(\frac{g \cos(\phi) \sin(2\Omega t) + 2 v_0 \Omega \cos(\theta) \sin(\phi) \cos(2\Omega t) - 2 v_0 \Omega \sin(\theta) \cos(\phi) \cos(2\Omega t) - 2 g \Omega \cos(\phi) t - 2 v_0 \Omega}{4\Omega^2}\right.$$

```
-->     expand(subst(φ = 0, sol_north_launch));
```

(% o31)

$$\left[x(t) = -\left(\frac{g \sin(2\Omega t)}{4\Omega^2}\right) + \frac{v_0 \sin(\theta) \cos(2\Omega t)}{2\Omega} + \frac{g t}{2\Omega} - \frac{v_0 \sin(\theta)}{2\Omega}, y(t) = v_0 \cos(\theta) t, z(t) = \frac{v_0 \sin(\theta) \sin(2\Omega t)}{2\Omega} + \frac{g}{\;}\right.$$

```
-->     expand(subst(φ = %pi / 2, sol_north_launch));
```

(% o32)

$$\left[x(t) = \frac{v_0 \cos(\theta)}{2\Omega} - \frac{v_0 \cos(\theta) \cos(2\Omega t)}{2\Omega}, y(t) = \frac{v_0 \cos(\theta) \sin(2\Omega t)}{2\Omega}, z(t) = v_0 \sin(\theta) t - \frac{g t^2}{2}\right]$$

```
-->     expand(subst(φ = - %pi / 2, sol_north_launch));
```

(% o33)

$$\left[x(t) = \frac{v_0 \cos(\theta) \cos(2\Omega t)}{2\Omega} - \frac{v_0 \cos(\theta)}{2\Omega}, y(t) = \frac{v_0 \cos(\theta) \sin(2\Omega t)}{2\Omega}, z(t) = v_0 \sin(\theta) t - \frac{g t^2}{2}\right]$$

```
-->     sol_west_launch: factor(subst([
        x_0 = 0,
        y_0 = 0,
        z_0 = 0,
        dx_0 = - v_0 * cos(θ),
        dy_0 = 0,
        dz_0 = v_0 * sin(θ)],
        sol));
```

(sol_ west_ launch)

$$[x(t) = -\left(\frac{g\cos(\phi)\sin(2\Omega t) + 2v_0\Omega\cos(\theta)\sin(2\Omega t) - 2v_0\Omega\sin(\theta)\cos(\phi)\cos(2\Omega t) - 2g\Omega\cos(\phi)t + 2v_0\Omega\sin(\theta)}{4\Omega^2}\right.$$

−> expand(subst(ϕ = 0, sol_west_launch));

(% o35)

$$[x(t) = -\left(\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\right) - \frac{g\sin(2\Omega t)}{4\Omega^2} + \frac{v_0\sin(\theta)\cos(2\Omega t)}{2\Omega} + \frac{gt}{2\Omega} - \frac{v_0\sin(\theta)}{2\Omega}, y(t) = 0, z(t) = \frac{v_0\sin(\theta)\,s}{2\Omega}$$

−> expand(subst(ϕ = %pi / 2, sol_west_launch));

(% o36)

$$\left[x(t) = -\left(\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\right), y(t) = \frac{v_0\cos(\theta)}{2\Omega} - \frac{v_0\cos(\theta)\cos(2\Omega t)}{2\Omega}, z(t) = v_0\sin(\theta)t - \frac{gt^2}{2}\right]$$

−> expand(subst(ϕ = - %pi / 2, sol_west_launch));

(% o37)

$$\left[x(t) = -\left(\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\right), y(t) = \frac{v_0\cos(\theta)\cos(2\Omega t)}{2\Omega} - \frac{v_0\cos(\theta)}{2\Omega}, z(t) = v_0\sin(\theta)t - \frac{gt^2}{2}\right]$$

```
−> sol_south_launch: factor(subst([
    x_0 = 0,
    y_0 = 0,
    z_0 = 0,
    dx_0 = 0,
    dy_0 = - v_0 * cos(θ),
    dz_0 = v_0 * sin(θ)],
    sol));
```

(sol_ south_ launch)

$$[x(t) = -\left(\frac{g\cos(\phi)\sin(2\Omega t) - 2v_0\Omega\cos(\theta)\sin(\phi)\cos(2\Omega t) - 2v_0\Omega\sin(\theta)\cos(\phi)\cos(2\Omega t) - 2g\Omega\cos(\phi)t + 2v_0\Omega}{4\Omega^2}\right.$$

−> expand(subst(ϕ = 0, sol_south_launch));

(% o39)

$$\left[x(t) = -\left(\frac{g\sin(2\Omega t)}{4\Omega^2}\right) + \frac{v_0\sin(\theta)\cos(2\Omega t)}{2\Omega} + \frac{gt}{2\Omega} - \frac{v_0\sin(\theta)}{2\Omega}, y(t) = -(v_0\cos(\theta)t), z(t) = \frac{v_0\sin(\theta)\sin(2\Omega t)}{2\Omega}\right.$$

−> expand(subst(ϕ = %pi / 2, sol_south_launch));

(% o40)

$$\left[x(t)=\frac{v_0\cos(\theta)\cos(2\Omega t)}{2\Omega}-\frac{v_0\cos(\theta)}{2\Omega},y(t)=-\left(\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\right),z(t)=v_0\sin(\theta)t-\frac{gt^2}{2}\right]$$

−> expand(subst(ϕ = - %pi / 2, sol_south_launch));

(% o41)
$$\left[x(t)=\frac{v_0\cos(\theta)}{2\Omega}-\frac{v_0\cos(\theta)\cos(2\Omega t)}{2\Omega},y(t)=-\left(\frac{v_0\cos(\theta)\sin(2\Omega t)}{2\Omega}\right),z(t)=v_0\sin(\theta)t-\frac{gt^2}{2}\right]$$

−> taylor(sin(2 * Ω * t), t, 0, 5);

(% o61/T/) $2\Omega t-\frac{4\Omega^3t^3}{3}+\frac{4\Omega^5t^5}{15}+...$

−> taylor(cos(2 * Ω * t), t, 0, 5);

(% o62/T/) $1-2\Omega^2t^2+\frac{2\Omega^4t^4}{3}+...$

```
−> expand(subst([
cos(2 * Ω * t) = 1-2*Ω^ 2*t^ 2+(2*Ω^ 4*t^ 4)/3,
sin(2 * Ω * t) = 2*Ω*t-(4*Ω^ 3*t^ 3)/3+(4*Ω^ 5*t^ 5)/15
], sol_tower));
```

(% o147)
$$\left[x(t)=\frac{g\Omega\cos(\phi)t^3}{3}-\frac{g\Omega^3\cos(\phi)t^5}{15},y(t)=-\left(\frac{g\Omega^2\cos(\phi)\sin(\phi)t^4}{6}\right),z(t)=-\left(\frac{g\Omega^2\sin(\phi)^2t^4}{6}\right)+\frac{g\Omega^2t^4}{6}-\frac{gt}{2}\right.$$

−> T2: rhs(solve(rhs(sol_tower[3]) = 0, t ^ 2)[1]);

(T2) $-\left(\frac{\left(g\sin(\phi)^2-g\right)\cos(2\Omega t)-g\sin(\phi)^2-4h\Omega^2+g}{2g\Omega^2\sin(\phi)^2}\right)$

−> T2: factor(subst(cos(2 * Ω * t) = 1-2*Ω^ 2*t^ 2+(2*Ω^ 4*t^ 4)/3, T));

(T2) $\frac{g\sin(\phi)^2t^2-gt^2+2h}{g\sin(\phi)^2}$

−> solve(T2 = t ^ 2, t)[2];

(% o160) $t=\frac{\sqrt{2}\sqrt{h}}{\sqrt{g}}$

--> D_east: expand(subst(t = sqrt(2 * h / g), rhs(sol_tower[1])));

(D_ east) $$\frac{\sqrt{g}\sqrt{h}\cos(\phi)}{\sqrt{2}\Omega} - \frac{g\sin\left(\frac{2^{\frac{3}{2}}\sqrt{h}\Omega}{\sqrt{g}}\right)\cos(\phi)}{4\Omega^2}$$

--> taylor(D_east, Ω, 0, 5);

(% o170/T/) $$\frac{\sqrt{2}^3\sqrt{h}^3\cos(\phi)\Omega}{3\sqrt{g}} - \frac{\sqrt{2}^5\sqrt{h}^5\cos(\phi)\Omega^3}{15\sqrt{g}^3} + \frac{\sqrt{2}^9\sqrt{h}^7\cos(\phi)\Omega^5}{315\sqrt{g}^5} + \ldots$$

--> D_north: expand(subst(t = sqrt(2 * h / g), rhs(sol_tower[2])));

(D_ north)
$$-\left(\frac{g\cos\left(\frac{2^{\frac{3}{2}}\sqrt{h}\Omega}{\sqrt{g}}\right)\cos(\phi)\sin(\phi)}{4\Omega^2}\right) + \frac{g\cos(\phi)\sin(\phi)}{4\Omega^2} - h\cos(\phi)\sin(\phi)$$

--> taylor(D_north, Ω, 0, 6);

(% o173/T/)
$$-\left(\frac{2h^2\cos(\phi)\sin(\phi)\Omega^2}{3g}\right) + \frac{8h^3\cos(\phi)\sin(\phi)\Omega^4}{45g^2} - \frac{8h^4\cos(\phi)\sin(\phi)\Omega^6}{315g^3} + \ldots$$

--> factor(D_north);

(% o177) $$-\left(\frac{\left(g\cos\left(\frac{2^{\frac{3}{2}}\sqrt{h}\Omega}{\sqrt{g}}\right) + 4h\Omega^2 - g\right)\cos(\phi)\sin(\phi)}{4\Omega^2}\right)$$

--> factor(sol_east_launch);

(% o179)
$$[x(t) = -\left(\frac{g\cos(\phi)\sin(2\Omega t) - 2v_0\Omega\cos(\theta)\sin(2\Omega t) - 2v_0\Omega\sin(\theta)\cos(\phi)\cos(2\Omega t) - 2g\Omega\cos(\phi)t + 2v_0\Omega\sin(\theta)}{4\Omega^2}\right.$$

--> taylor(rhs(sol_east_launch[1]), Ω, 0, 2);

(% o197/T/)
$$v_0\cos(\theta)t + \frac{\left(g\cos(\phi)t^3 - 3v_0\sin(\theta)\cos(\phi)t^2\right)\Omega}{3} - \frac{2v_0\cos(\theta)t^3\Omega^2}{3} + \ldots$$

−> taylor(rhs(sol_east_launch[2]), Ω, 0, 2);

(% o198/T/)

$$-\left(v_0\cos(\theta)\sin(\phi)t^2\Omega\right)-\frac{\left(g\cos(\phi)\sin(\phi)t^4-4v_0\sin(\theta)\cos(\phi)\sin(\phi)t^3\right)\Omega^2}{6}+\ldots$$

−> taylor(rhs(sol_east_launch[3]), Ω, 0, 2);

(% o199/T/)

$$-\left(\frac{gt^2-2v_0\sin(\theta)t}{2}\right)+v_0\cos(\theta)\cos(\phi)t^2\Omega-\frac{\left(\left(g\sin(\phi)^2-g\right)t^4+\left(-\left(4v_0\sin(\theta)\sin(\phi)^2\right)+4v_0\sin(\theta)\right)t^3\right)\Omega}{6}$$

−> factor(sol_north_launch);

(% o200)

$$[\,x(t)=-\left(\frac{g\cos(\phi)\sin(2\Omega t)+2v_0\Omega\cos(\theta)\sin(\phi)\cos(2\Omega t)-2v_0\Omega\sin(\theta)\cos(\phi)\cos(2\Omega t)-2g\Omega\cos(\phi)t-2v_0\Omega}{4\Omega^2}\right.$$

−> taylor(rhs(sol_north_launch[1]), Ω, 0, 3);

(% o212/T/)

$$\frac{\left(g\cos(\phi)t^3+(3v_0\cos(\theta)\sin(\phi)-3v_0\sin(\theta)\cos(\phi))\,t^2\right)\Omega}{3}-\frac{\left(g\cos(\phi)t^5+(5v_0\cos(\theta)\sin(\phi)-5v_0\sin(\theta)\cos(\phi\right.}{15}$$

−> taylor(rhs(sol_north_launch[2]), Ω, 0, 3);

(% o213/T/)

$$v_0\cos(\theta)t-\frac{\left(g\cos(\phi)\sin(\phi)t^4+\left(-(4v_0\sin(\theta)\cos(\phi)\sin(\phi))-4v_0\cos(\theta)\cos(\phi)^2+4v_0\cos(\theta)\right)t^3\right)\Omega^2}{6}+\ldots$$

−> taylor(rhs(sol_north_launch[3]), Ω, 0, 3);

(% o214/T/)

$$-\left(\frac{gt^2-2v_0\sin(\theta)t}{2}\right)-\frac{\left(\left(g\sin(\phi)^2-g\right)t^4+\left(-\left(4v_0\sin(\theta)\sin(\phi)^2\right)-4v_0\cos(\theta)\cos(\phi)\sin(\phi)+4v_0\sin(\theta)\right)\right.}{6}$$